

Chapter 101 — Surgical, Radiographic, and Endoscopic Anatomy of the Female Pelvis

IF YOU READ NOTHING ELSE

- **Ureter under uterine artery** ("water under the bridge") → hysterectomy injury site.
- **Sacrospinous ligament (SSL) fixation** → risks **pudendal nerve** (near ischial spine) and **sacral plexus** (just posterior).
- **Sacrocolpopexy sutures** → **anterior longitudinal ligament (ALL) over S1**, not L5/S1 disc.
- **Presacral venous plexus**: valveless; 1–2 mm vein → >1 L/min bleeding.
- **Lithotomy foot drop** → **common peroneal nerve** at fibular head.
- **Distal ⅓ urethra** → **inguinal nodes**; proximal ⅔ → iliac and obturator.
- **Left ovarian vein** → **left renal vein**; right → inferior vena cava (IVC).
- **Bladder dome injury** → **intraperitoneal** → **operative repair**.
- **T2-weighted MRI** = **best test for urethral diverticulum**.
- **Internal iliac ligation tolerated** (collaterals); **external iliac ligation** → **limb emergency**.

TRANSVAGINAL APPROACH

- **Vulva** = mons, labia majora and minora, clitoris, vestibule.
- **Meatus 1 cm anterior** to vaginal opening.
- **Skene's glands** = prostate homologue; ventral to meatus.
- **Bartholin's gland** under posterior vestibular bulb → cyst or abscess.
- **Crura** → **ischiocavernosus**; **bulbs** → **bulbospongiosus**.
- **Perineal membrane** = old "urogenital diaphragm"; pierced by urethra and vagina.
- **Deep space**: striated sphincters, deep transverse perinei. **Superficial**: bulbs, clitoris, Bartholin's.
- **Colles fascia** = female dartos; roof of superficial space.
- **Perineal body** = bulbospongiosus + external anal sphincter + superficial and deep transverse perinei.
- **Episiotomy angled laterally** → spares external anal sphincter.
- **Female pelvis**: oval inlet (male heart-shaped); **obtuse** subpubic angle; short wide sacrum.
- **Standing**: ASIS and symphysis vertical; inlet faces anteriorly → load onto bone.
- **Ischial spine** ≈ **S5**; landmark for apical suspension.
- **Interspinous line** divides true pelvis anterior vs posterior compartments.
- **True vs false pelvis**: below vs above pelvic brim (promontory → symphysis).
- **Sacral promontory neurovascular bed**: avoid in mesh fixation.
- **S1 foramina fan lateral to S4** → finding S3 for neuromodulation.
- **SSL**: ischial spine → lower sacrum and coccyx.
- **Greater sciatic foramen**: piriformis, sacral plexus. **Lesser**: pudendal nerve and vessels, obturator internus.
- **Cooper's ligament** → Burch urethropexy anchor.
- **Obturator canal superolateral** → transobturator trocar: groin pain, bleeding.
- **Levator ani** = puborectalis + pubococcygeus + iliococcygeus; **S3–S5 (mainly S4)**.
- **Puborectalis** = anorectal sling; **pubococcygeus** = pubovaginal (urethrovaginal) sling.
- **Wide levator hiatus** → **prolapse**.
- **Levator plate** = horizontal visceral shelf; loses orientation with lost tone.
- **Coccygeus**: ischial spine → sacrum and coccyx, on the SSL.
- **Arcus tendineus**: ischial spine → pubis.
- **Martius flap**: labial fat; dual supply — external pudendal (femoral) + internal pudendal.
- **Sacral plexus L4–S4**. **Pudendal S2–S4** → Alcock's canal.
- **Pudendal block**: SSL near ischial spine, transvaginal.

- **Pudendal branches**: inferior rectal, perineal, dorsal nerve of clitoris.
- **Pelvic floor**: direct S3–S5; pelvic somatic nerves (L4–S4) → levator and striated urethral sphincter.
- **Ilioinguinal injury**: hernia repair, appendicectomy, caesarean, abdominal hysterectomy.
- **Genitofemoral injury**: herniorrhaphy, lymphadenectomy.
- **Vulval lymph**: superficial inguinal → deep inguinal → pelvic.
- **Ilioinguinal L1, genitofemoral L1–L2**: mons and labia.
- **Posterior femoral cutaneous S1–S3**: perineum.
- **Post-lithotomy leg pain and swelling** → **compartment syndrome**: surgical emergency.
- **Fascia lata harvest**: avoid near lateral condyle (narrow band).

TRANSABDOMINAL APPROACH

Cut-offs:

- **Urethra 4.7–5.1 cm**; cervix 2.5 cm; tubes 10–12 cm; rectum 12–15 cm.
- **Vagina: anterior 7.5 cm vs posterior 9 cm**.
- **Pelvic ureter** ≈ **half** ureteric length.
- **Aorta: T12** → **bifurcates L4** (left anterolateral).
- **Common iliac artery: right** ≈ 5 cm, **left** ≈ 4 cm.
- **Common iliac veins unite** ≈ **L5**.
- **ALL 1.3–2.5 mm thick**.
- **Mid-promontory** → **right ureter and left common iliac vein** ≈ 2.7 cm each.

Urethra and bladder:

- **Internal sphincter**: smooth, involuntary, bladder neck. **External**: striated, voluntary, distal ⅓.
- **Closure pressure** = **mucosa + submucosa**; **oestrogen-dependent**.
- **Diverticulum** = infected or ruptured periurethral gland through periurethral fascia.
- **Ventral** = vaginal side; **dorsal** = pubic side.
- **Pubovaginal sling** → **bladder neck**; **mid-urethral sling** → **mid-urethra**.
- **Bladder base injury** → **vesicovaginal or vesicouterine fistula**.
- **Striated sphincter complex**: urethral compressor (closes on vaginal wall) + urethrovaginal sphincter (narrows hiatus).
- **Proximal urethra + bladder neck** = female internal sphincter.
- **Epithelium**: transitional → squamous distally.
- **Uterus anteverted, anteflexed**; posterior fornix deeper than anterior.
- **Tube**: interstitial → isthmus → ampulla → infundibulum.
- **Ovarian fossa**: obliterated umbilical artery, ureter, internal iliac artery.
- **Vesicovaginal space**: divide bladder–vaginal smooth muscle fibres (fistula repair).

- **Rectum:** retroperitoneal, no taeniae or haustra.

Ureter danger points:

- Pelvic brim over **common iliac bifurcation**.
- **Under uterine artery;** beside cardinal ligament near cervix.
- **Anterior vaginal fornix** (transvaginal surgery).
- Lateral to **uterosacral ligament, closest at cervix**.
- **Left ureter:** base of sigmoid mesocolon; landmark **intersigmoid fossa**.

Ligaments:

- **Infundibulopelvic:** ovary → side wall; carries ovarian vessels.
- **Ovarian ligament:** ovary → uterus.
- **Round ligament** = gubernaculum homologue; ends in labium majus.
- **Cardinal (Mackenrodt's):** side wall → cervix; uterine vessels.
- **Uterosacral:** sacrum S2–S4 → cervix; apical suspension anchor.
- **Pubourethral** = puboprosthetic homologue; weakness → stress urinary incontinence (SUI).
- **Vesicopelvic fascia defect** → **lateral cystocele**.
- **Most support** = **intermediate fascial stratum** (parametrium, paracolpium).
- **Cardinal and uterosacral carry pelvic plexus** → hysterectomy → bladder dysfunction.
- **Uterosacral suspension** → sacral nerve (S1–S4) entrapment.

Vessels and nodes:

- **Internal iliac divides** at upper border of greater sciatic foramen.
- **Posterior division** = **parietal only:** iliolumbar, lateral sacral, superior gluteal.
- **Middle sacral artery** behind left common iliac vein.
- **Obturator and uterine artery origins variable**.
- **Uterine collaterals:** contralateral uterine + ovarian (aorta, below renals).
- **Internal iliac vein posteromedial** to artery; often duplicated.
- **External iliac vein** = femoral vein continuation behind inguinal ligament.
- **Most visceral lymph** → **internal iliac** (presacral, obturator, internal pudendal).
- **Bladder and vagina** → **middle external iliac chain**.
- **Upper ⅔ vagina** → **pelvic;** lower ⅓ and vulva → **inguinal**.
- **Ovary and tube** → **along gonadal vessels** (para-aortic).
- **Common iliac nodes** → **lateral aortic nodes**.
- **Leissner levels:** 1 iliac, obturator, perivesical → 2 common iliac, presacral → 3 paracaval, para-aortic, interaortocaval.

ENDOSCOPIC ANATOMY

- **30° lens at bladder neck** → orthotopic orifices on trigonal ridge.
- **Patulous orifice** → reflux; ectopic → duplication.
- **Prolapse obscuring orifices** → **vaginal sponge stick**.
- **Impressions:** symphysis anterior; bowel lateral; uterus posterior.
- **Squamous metaplasia, cystitis cystica:** biopsy if in doubt.
- **Post-radiation mucosa:** friable, hypervascular.
- **Urethroscopy:** coaptation at mid-urethra and bladder neck; look for mesh, stones, diverticular ostia.

RADIOGRAPHY

- **Translabial ultrasound:** mesh, slings, bulking agents; bladder neck mobility on Valsalva.
- **Endoanal ultrasound:** **internal sphincter defect** = **echogenic break;** **external** = **hypoechoic lesion**.
- **MRI** = most comprehensive radiation-free pelvic imaging.

- **Dynamic pelvic floor MRI:** T2 HASTE or SSFSE (fast single-shot sequences).
- **Uterine T2:** endometrium high, junctional zone low, myometrium intermediate–high.
- **Benign vaginal cysts:** müllerian, epidermal inclusion, Gartner duct, Bartholin, Skene.
- **Pubococcygeal line:** inferior symphysis → last coccygeal joint.
- **MR defecography:** descent below line → mild, moderate or severe per compartment.
- **MR defecography if defaecatory dysfunction or symptoms** > examination findings.

CLASSIC TRAPS

- **Trap:** infundibulopelvic ligament joins ovary to uterus — **Actually:** ovary to side wall; ovarian ligament joins uterus.
- **Trap:** lithotomy foot drop is a pelvic nerve injury — **Actually:** common peroneal compression at fibular head.
- **Trap:** both ovarian veins drain to IVC — **Actually:** left → left renal vein.
- **Trap:** whole urethra drains to pelvic nodes — **Actually:** distal ⅓ → inguinal.
- **Trap:** posterior internal iliac division supplies viscera — **Actually:** parietal branches only.
- **Trap:** pubovaginal sling sits mid-urethra — **Actually:** bladder neck; synthetic sling mid-urethra.
- **Trap:** sacrocolpopexy sutures into L5/S1 disc — **Actually:** ALL over S1 (discitis risk).
- **Trap:** dome injury heals with catheter — **Actually:** intraperitoneal → repair.
- **Trap:** internal sphincter tear is hypoechoic — **Actually:** echogenic break; external tear hypoechoic.
- **Trap:** coccygeus is part of levator ani — **Actually:** separate muscle on the SSL.